Study A — Spatial Preservation vs. Parameter Count

A controlled architecture ablation for WiFi indoor localization

DLoc lightweight localization — E-JUST graduation project

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Bottom line. Across a controlled ladder of nine architectures on the WILD Jacobs-Hall in-distribution split, *adding parameters does not improve accuracy, but adding spatial structure does*. Pouring capacity into the decoder of the plain MobileNetV2 leaves the error essentially flat ($0.881 \to 0.883$ m as parameters rise $2.02 \to 2.92$ M), whereas adding U-Net skip connections lowers it at *every* capacity, and a skip model with *fewer* parameters (2.41 M) beats a no-skip model with *more* (2.92 M). Multi-scale features (ASPP) help further (best mean 0.791 m). The skip vs. no-skip difference is highly significant (paired Wilcoxon $p = 1.8 \times 10^{-11}$). **This directly supports the thesis** and hardens the Mamba / 2-D spatial argument.

1 Objective

This study answers the question posed by the supervisor: isolate the contribution of each architectural ingredient on top of a MobileNetV2 backbone — +decoder → +skip connections → +multi-scale features → +bottleneck widths → +decoder capacities — and test whether *spatial preservation contributes independently of parameter count*. A positive result strengthens the central argument of the project: that *preserving 2-D spatial structure*, not raw model size, is what makes a compressed CSI localizer accurate (the same reason the 1-D Mamba variant, which flattens the 2-D heatmap, fails).

2 Experimental design

A single *configurable* student (`student_configurable.py`) exposes four knobs — `use_skips`, `decoder_width`, `width_mult` (encoder/bottleneck width), and `bottleneck` (`none`/`aspp`) — and reproduces the shipped networks exactly (no-skip, width 1.0 = the 2.27 M MobileNetV2; skip, width 1.0 = the 2.41 M MobileNetV2-UNet). Thirteen training runs cover nine configurations.

- **Capacity axis = decoder width** (0.5/1.0/2.0). The encoder stays ImageNet-pretrained at width 1.0, so this axis varies capacity *without* a pretraining confound. Encoder-width (`width_mult` 0.5/1.4) is a secondary, *from-scratch* axis (no pretraining at non-unit width) and is reported with that caveat.
- **Data:** WILD Jacobs-Hall in-distribution split (Fig. 10b): 17,574 training / 4,260 test samples. Input $4 \times 161 \times 361$ AoA–ToF heatmaps; metric is the Euclidean peak-to-peak distance $\times 0.1$ m.
- **Training:** Adam, lr 10^{-4}, batch 32, 120 epochs, cosine annealing; MSE loss. Best checkpoint selected by *mean* error (the primary metric; the median is pixel-quantized to $0.1\sqrt{n}$ and cannot resolve sub-decimetre differences).
- **Seeds:** the two headline configurations (no-skip and skip at decoder-width 1.0) use 3 seeds

(42/123/777) for error bars; the remaining configurations are single-seed (trend only). Statistical test: paired Wilcoxon on per-sample errors (seed 42, $n = 4260$).

3 Results

Table 1: All nine configurations, ordered by parameter count. Mean error is the primary metric (± sample std over 3 seeds for the two anchors). "pre" = ImageNet-pretrained encoder. Best mean in bold.

Config	Skips	Multi-scale	Pre	Params	FLOPs	Mean (m)	P90 (m)
skip, enc-w 0.5†	yes	–	no	0.83M	0.76G	1.037	1.868
no-skip, dec-w 0.5	no	–	yes	2.02M	0.92G	0.881	1.565
skip, dec-w 0.5	yes	–	yes	2.09M	1.00G	0.851	1.513
no-skip, dec-w 1.0	no	–	yes	2.27M	1.28G	0.882 ± 0.018	1.570
skip, dec-w 1.0	yes	–	yes	2.41M	1.45G	0.830 ± 0.003	1.435
no-skip, dec-w 2.0	no	–	yes	2.92M	2.51G	0.883	1.562
skip, dec-w 2.0	yes	–	yes	3.19M	2.85G	0.866	1.513
skip, dec-w 1.0, ASPP	yes	yes	yes	3.28M	n/a‡	**0.791**	**1.432**
skip, enc-w 1.4†	yes	–	no	4.31M	2.28G	0.855	1.581

† Encoder-width ≠ 1.0: trained from scratch (no ImageNet pretraining) — confounded, secondary axis. ‡ The FLOP counter does not support the ASPP module; value omitted.

Table 2: Mean error vs. capacity, read down each column (pretrained, decoder-width axis). The no-skip row is *flat* as parameters grow; the skip row is lower at every width.

Decoder width	0.5	1.0	2.0
no-skip (params →)	2.02M	2.27M	2.92M
mean error (m)	0.881	0.882	0.883 *(flat)*
skip (params →)	2.09M	2.41M	3.19M
mean error (m)	0.851	**0.830**	0.866 *(lower everywhere)*

4 Findings

1. **Skip connections help, significantly.** At matched decoder-width 1.0, skips lower the mean from 0.882 ± 0.018 to 0.830 ± 0.003 m (-6.0%); the per-sample paired Wilcoxon test gives $p = 1.8 \times 10^{-11}$ ($n = 4260$). The tiny std of the skip anchor (± 0.003) shows the effect is not seed noise.

2. **Parameter count alone buys nothing (the key result).** The no-skip mean is *flat* — 0.881/0.882/0.883 m — as the decoder grows 2.02 → 2.27 → 2.92 M parameters (and FLOPs nearly triple, 0.92 → 2.51 G). Tripling the decoder's capacity does not move the accuracy at all.

3. **Spatial preservation beats extra parameters directly.** A skip model with *fewer* parameters outperforms a no-skip model with *more*: skip dec-w 0.5 (2.09 M, 0.851 m) < no-skip dec-w 1.0 (2.27 M, 0.882 m), and skip dec-w 1.0 (2.41 M, 0.830 m) < no-skip dec-w 2.0 (2.92 M, 0.883 m). Interpolating the no-skip curve to equal parameter counts, skips are better by 0.031 m (at 2.09 M) and 0.053 m (at 2.41 M).

4. **Multi-scale features help further.** Adding an ASPP bottleneck to the skip model gives

the best result of all (mean 0.791 m, P90 1.432 m) — another spatial, not capacity-driven, gain.

5. **Widening the encoder/bottleneck does not help.** The widest model (encoder-width 1.4, 4.31 M — 79% more parameters than the 2.41 M skip model) is *no better* (0.855 vs 0.830 m). This is consistent with (2), though it is also confounded by being trained from scratch, so we treat it as supporting rather than decisive.

5 Does it help the thesis? — Yes

1. **It justifies the chosen architecture.** The thesis selects the MobileNetV2-UNet as the final model because skip connections improve accuracy. Study A turns that choice from an empirical pick into a *controlled, statistically significant* claim: skips are the active ingredient, and the gain is not an artifact of the extra parameters they add.

2. **It hardens the Mamba / 2-D spatial argument.** The thesis explains Mamba's failure (10.56 m) as an *architectural mismatch*: flattening the 2-D heatmap into a 1-D sequence destroys spatial locality. Study A provides the positive counterpart — when spatial structure is *preserved and enriched* (skips, then multi-scale), accuracy improves, while merely adding capacity does nothing. The failure of Mamba is therefore a *spatial-information* problem, not a capacity problem — exactly the thesis's claim, now backed by a parameter-controlled ablation.

3. **It points to a concrete future direction.** ASPP (multi-scale) gives the best result here, suggesting richer multi-scale spatial aggregation as the next architectural step — a well-supported "future work" item.

In short, Study A supplies the "*why*" behind the headline result: compressed CSI localization is driven by spatial-structure preservation, not model size.

6 Caveats (for honesty in the write-up)

- **Absolute level.** These means (e.g. 0.830 m for the skip dec-w 1.0 model) are a touch higher than the headline paper's 0.640 m *median*, because (i) selection is by *mean* (the median is pixel-quantized and cannot resolve the sub-decimetre gaps this study measures) and (ii) this is the configurable re-implementation. The study's value is in the *relative* ladder comparisons, which are internally consistent.
- **Single-seed configs.** Only the two decoder-width-1.0 anchors are multi-seed; the other seven configurations are single runs (direction/trend, not error bars). The decoder-width axis trend (flat no-skip, lower skip) is consistent across its points, and the one multi-seed pair is tight, which supports the conclusion.
- **From-scratch secondary axis.** The encoder-width points (0.5, 1.4) lack ImageNet pretraining and are reported only as a caveated secondary axis; the clean capacity axis is decoder-width, where all points are pretrained.

7 Conclusion

On a controlled nine-architecture ladder, **spatial preservation — not parameter count — drives WiFi-localization accuracy**. Adding decoder capacity to the plain MobileNetV2 leaves the error flat ($\sim$0.88 m across a 1.4$\times$ parameter range), while skip connections lower it at every capacity and even outperform larger no-skip models; multi-scale (ASPP) lowers it further to 0.791 m. The skip-vs-no-skip gain is statistically significant ($p = 1.8 \times 10^{-11}$) and seed-stable.

This is exactly the result the supervisor asked for: *spatial preservation contributes independently of parameter count*. It validates the thesis's architecture choice, converts the Mamba failure into a principled spatial-information argument, and motivates multi-scale spatial aggregation as future work.